25 2026-04-06 | Week 12 | Lecture 25

Based on chapter 5.4

25.1 The central limit theorem

Theorem 115 (Central limit theorem)**.** *Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance $\sigma^2 > 0$. If n is sufficiently large, then $\overline{X}$ has approximately a normal distribution with mean $\mu_{\overline{X}} = \mu$ and variance $\sigma^2_{\overline{X}} = \frac{\sigma^2}{n}$. The larger n, the better the approximation.*

In symbols,

$$\overline{X} \approx \underbrace{\mu + \frac{\sigma}{\sqrt{n}} Z}_{\mathcal{N}(\mu, \frac{\sigma^2}{n})} \tag{15}$$

where $Z \sim \mathcal{N}(0,1)$.

Usually $n \geq 30$ is big enough for the approximation to be valid.

Corollary 116. *The distribution of the standardardized sample mean is approximately a standard normal; that is,*

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \approx Z, \tag{16}$$

where $Z \sim \mathcal{N}(0,1)$.

Example 117 (Witch's brew - similar to example 5.27 in textbook)**.** A witch decides to make 50 batches of potions. Unfortunately, her hut is not very clean and bugs keep crawling into the brew. Suppose that the amount of bugs contaminating the ith batch is a random variable X_i with mean $\mu = 4g$ and standard deviation $\sigma = 1.5g$. Assume the witch prepares her 50 batches independently. Let

$$\overline{X} = \frac{X_1 + \ldots + X_{50}}{50}$$

be the average amount of impurity.

Question: What is the probability that $3.5 \leq \overline{X} \leq 3.8$?
Solution: Here, $n = 50 > 30$ so we can apply the central limit theorem. By the central limit theorem, $\overline{X}$ has distribution approximately

$$\begin{aligned} \overline{X} &\approx \mu + \frac{\sigma}{\sqrt{n}} Z \\ &= 4 + \frac{3}{2\sqrt{50}} Z, \end{aligned}$$

where $Z \sim \mathcal{N}(0,1)$ is a standard normal random variable. Therefore

$$\begin{aligned} \mathbb{P}\left[3.5 \leq \overline{X} \leq 3.8\right] &\approx \mathbb{P}\left[3.5 \leq 4 + \frac{3}{2\sqrt{50}} Z \leq 3.8\right] \\ &= \mathbb{P}\left[-\frac{1}{2} \leq \frac{3}{2\sqrt{50}} Z \leq -\frac{1}{5}\right] \\ &= \mathbb{P}\left[-\frac{\sqrt{50}}{3} \leq Z \leq -\frac{2\sqrt{50}}{15}\right] \\ &= \mathbb{P}\left[-2.36 \leq Z \leq -.94\right] \\ &= \int_{-2.36}^{-.94} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy \\ &\approx .16 \end{aligned}$$

End of Example 117. □

Example 118 (CLT Coin Flipping)**.** Flip a fair coin 100 times. What is the probability that you get at least 55 heads?

Solution: Let

$$X_i = \begin{cases} 0 & : & i^{th} \text{ coin flip is tails} \\ 1 & : & i^{th} \text{ coin flip is heads} \end{cases}$$

We want to compute $\mathbb{P}\left[X_1 + \ldots + X_n \geq 55\right]$, or equivalently,

$$\mathbb{P}\left[\overline{X} \geq .55\right].$$

We will use Eq. (16). First observe that

$$\mu = \mathbb{E}\left[X_i\right] = \frac{1}{2}$$

and that

$$\sigma = \sqrt{\operatorname{Var}(X_i)} = \frac{1}{2}$$

(you should check these equations).

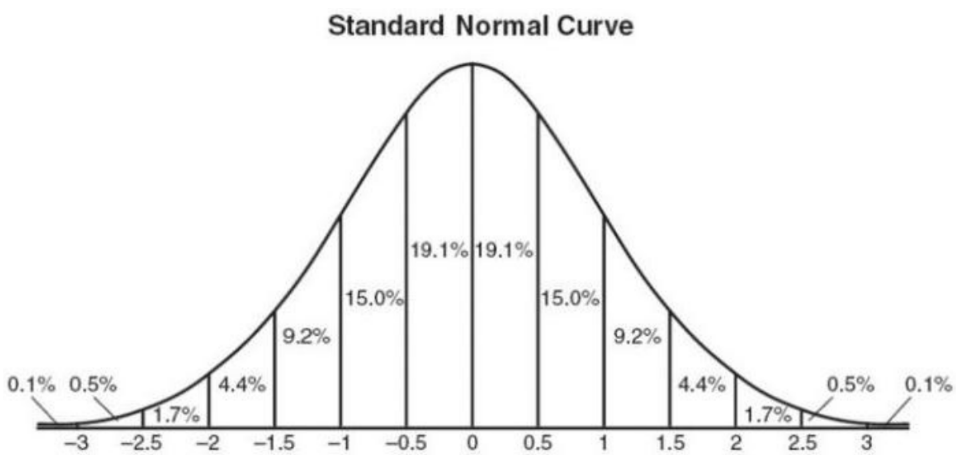
Therefore

$$\begin{aligned} \mathbb{P}\left[\overline{X} \geq .55\right] &= \mathbb{P}\left[\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{.55 - \mu}{\sigma/\sqrt{n}}\right] \\ &\approx \mathbb{P}\left[Z \geq \frac{.55 - \mu}{\sigma/\sqrt{n}}\right] && \text{by CLT (Eq. (16))}\,. \end{aligned}$$

The above is the key step. Plugging in $n = 100$, $\mu = \frac{1}{2}$ and $\sigma = \frac{1}{2}$ gives

$$\mathbb{P}\left[\overline{X} \geq .55\right] = \mathbb{P}\left[Z \geq 1\right] \approx .16,$$

where we get .16 by consulting the following figure:



End of Example 118. □

Remark 119. When reviewing the key step in Example 118, it is helpful to recall the standardization transformation defined in Remark 94. The central limit theorem says that, approximately, $\overline{X} \sim \mathcal{N}(\mu, \sigma^2/n)$. Hence, by Remark 94,

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \approx Z$$

where $Z \sim \mathcal{N}(0, 1)$. This is what justifies the key step in Example 118.

25.2 Point estimation

Section 6.1

Recall a **population** is a large body of data that is the target of our interest, and a **statistic** is any quantity whose value can be computed from a sample.

Often, we have some quantity of interest, called a ***target parameter***, and we want a single "best guess" of some quantity of interest. When this is the case, the we call a statistic an estimator:

Definition 120 (Estimator)**.** An ***estimator*** is a statistic, that is a function $T(X_1, \ldots, X_n)$ of a sample, that is used to approximate a target parameter. An ***estimate*** is the realized value of an estimator (e.g., a number) that is obtained when the sample is actually taken.

In words, an estimator is a rule, often expressed as a formula, that tells us how to calculate the value of an estimate based on the measurements contained in a sample.